Weekend Ms

28 August 2026

Contents

Submission draft. The full working paper, an electronic companion with all supporting tables, and code reproducing every figure are available at the project repository.

Abstract

Asset prices are far more volatile when exchanges are open than when they are shut. Distinguishing information arrival from the mechanical effect of closure has never been possible in equity markets, where "the exchange is closed" and "no one is trading" are the same event. Cryptocurrency separates them: Deribit trades continuously while the traditional financial system does not. We show that Bitcoin realized variance is 42% lower on weekends despite uninterrupted trading, with Saturday the quietest day of the week, and that the effect is near-identical across four underlyings with different settlement conventions (34%–42%). It roughly doubles, to 65%, for tokenized gold, which trades on the same venue while gold's own market is shut all weekend. Because Deribit lists daily expiries on all seven weekdays, contracts quoted at the same instant differ in how much weekend calendar time they span, which identifies the market's implied weekend discount from purely within-instant variation. The market prices roughly seven eighths of the effect in the mature books. Its one demonstrable failure is *within* the weekend: all four books price Saturday as indistinguishable from Sunday when Saturday is reliably quieter — and the venue's own expiry schedule makes that error unarbitrageable, offering both sides of the offsetting spread in 13 hours of the week and none of them on a weekday.

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1. Introduction

French and Roll (1986) established that US equity variance per calendar hour collapses when the exchange is shut, and set out three candidate explanations: public information arrives disproportionately during business hours; private information is impounded when informed investors trade;

or trading itself generates pricing errors. Distinguishing among them has proved difficult for a structural reason. In a conventional market, closure and the absence of trading are the same event, and both coincide with the hours in which most public information is released. The three explanations are confounded by construction.

Cryptocurrency markets break that confound. Deribit trades options and perpetual futures continuously, including weekends, while the traditional financial system does not. Trading is possible at all times, so any weekend variance effect cannot be attributed to a venue being shut.

Weekends are quiet even though the market is open. Realized variance is 42% lower on Saturdays and Sundays than on weekdays for Bitcoin, 39% for Ether, 38% for XRP and 34% for Solana, with Saturday the quietest day in all four. The natural reading is that much of what moves crypto prices originates in traditional market hours — macroeconomic releases, institutional flow, the spot-ETF complex — and the fact that four assets with different settlement conventions, holder bases and listing histories inherit the *same* weekly rhythm is what makes that reading hard to avoid. A fifth asset sharpens it. PAXG, tokenized gold, trades continuously on the same venue while gold's own market in London and on COMEX is shut all weekend; its weekend variance is **65% below weekday**, roughly double the crypto effect. Turn the traditional calendar up and the weekend deepens, on the same exchange with the same estimator.

Options price most of the effect. Deribit lists daily expiries on all seven weekdays, so contracts quoted at the same instant differ in how much weekend calendar time they span: on a Thursday, a Friday expiry covers no weekend while a Monday expiry covers most of one. Regressing squared implied volatility on that fraction, with day fixed effects absorbing the volatility level and every other common shock, identifies the implied weekend discount from within-instant variation alone. This design has no equity analogue — there are no Saturday expiries when the exchange is closed. Bitcoin's options price 85.0% of Bitcoin's own realized weekend effect and Ether's 85.5% of Ether's. None of the four residual pricing errors is individually significant, and we do not build on them.

The failure is within the weekend, not across assets. In all four books the market prices Saturday as statistically indistinguishable from Sunday, when Saturday is reliably the quieter of the two. Across the rest of the week the implied day-of-week profile tracks the realized one closely — correlations of +0.98, +0.88, +0.89 and +0.82 — so the market reads the calendar well everywhere except here, where it resolves "the weekend" as a single undifferentiated block. Four independent books make the same mistake in the same place.

We then show why that particular error survives, and the answer is a property of the contract calendar rather than of beliefs. The trade that would correct it — sell the Saturday-heavy contract, buy the Sunday-heavy one — requires both to exist at the same moment. Enumerating every hour of the week against every listed expiry, they coexist in **13 hours out of 168, none of them on a weekday**. Monday through Friday no Sunday-heavy contract is listed at all, because any contract reaching Sunday must pass through Saturday first. The market's one demonstrably wrong ranking of calendar time is the one its own expiry schedule prevents anyone from arbitraging.

Two further results bound the interpretation. Weekend returns do carry fatter tails, so there is a weekend-specific risk to charge for — but decomposing realized variance into continuous and jump components bounds what a jump premium can price, and every book's quote falls outside that bound; two of the four would need a *negative* price of jump risk, since their quotes sit below their own realized weekend variance. And the residual error is not harvestable: a vega-matched calendar spread isolating it earns +0.042 per unit vega gross against 0.066 of measured fees and spread, so it has never cleared zero at taker cost.

Contribution. The paper's design point is that a single venue supplies both halves of the test. The realized side compares weekend to weekday variance on an exchange that never closes, so closure is held fixed while the information environment varies; the reference asset raises the dose by holding the venue fixed while shutting the underlying's own market. The implied side compares contracts quoted in the same book at the same instant that differ only in how much weekend they span, which is possible because expiries are listed on all seven weekdays. Neither comparison is available in equity or listed-futures markets, where closure, non-trading and the news calendar move together and where no contract can expire while the exchange is shut.

What that buys is a measurement of how a derivatives market prices a known, mechanical, repeating feature of calendar time — and, because the same venue supplies the realized benchmark, a measurement of how well. The answer is "substantially, but at the wrong resolution and against the wrong moment," and both failures are identified rather than inferred.

Related literature

French and Roll (1986) is the origin of the question. Barclay, Litzenberger and Warner (1990) came closest to separating the mechanisms, using a natural experiment in the opposite direction from ours: when the Tokyo Stock Exchange opened on Saturdays, weekend variance rose while weekly variance was unchanged despite higher volume. They varied whether an exchange was *open* while holding the information environment roughly fixed; we hold the exchange open always and vary whether the traditional financial system is running. The two designs bracket the question from opposite sides, and both point to trading time rather than calendar time as the relevant clock.

On the option side, practitioners have long used trading-day rather than calendar-day clocks, and the academic treatment of weekend and holiday effects in implied volatility is thin relative to its practical importance. Our contribution is a venue where the clock convention is identified from within-instant variation rather than assumed.

2. Institutional setting and data

2.1 Why Deribit

Deribit is the dominant venue for crypto options and, critically for this paper, lists **daily expiries on all seven weekdays**. Across 342,827 instruments in the four books used here the expiry-weekday distribution is close to uniform apart from the expected Friday concentration — every underlying lists between 12% and 14% of its instruments on each of the six non-Friday weekdays. From 2020 onward there is an expiry on essentially every calendar date (365 of 365 in each year from 2021 to 2025), and short-dated contracts are the norm rather than a curiosity: roughly 80% of instruments have seven days or less of life.

That schedule is what makes the design work. Because Saturday and Sunday expiries exist and trade, a contract quoted on a Thursday afternoon may span no weekend at all or most of one, and the two sit side by side in the same book at the same instant. No equity or listed-futures market offers this: when the exchange is shut there is nothing to expire into, so weekend coverage is perfectly collinear with maturity.

2.2 Sample

We use the complete public trade history collected from `history.deribit.com`: 24,349,954 Bitcoin trades from November 2016, 16,207,332 Ether trades from March 2019, 695,295 Solana trades and 234,640 XRP trades from early 2024, running to August 2026 — 41.5 million trades in all. Each carries the exchange's own implied volatility, the index level, the aggressor side, and flags for block trades, combinations and liquidations.

The baseline sample excludes block trades and combinations, which are bilaterally negotiated or double-counted, retains liquidations, and keeps contracts with $|\Delta|$ between 0.30 and 0.70 and maturity between six hours and fourteen days. Realized variance comes from five-minute perpetualfuture returns over 2,923 Bitcoin days, 2,704 Ether days, 915 Solana days and 887 XRP days, at 100%, 100%, 99.9% and 99.9% bar completeness. Returns spanning a feed gap are dropped rather than winsorized, since a multi-period return carrying a one-period label inflates exactly the tail statistics §6 depends on.

Solana and XRP options are USDC-settled (linear) rather than coin-settled, carry contract sizes of ten and one thousand rather than one, and trade on separate instrument families. They require their own premium and hedging conventions throughout, which is what makes them genuinely independent replications rather than relabellings of the same book. One caveat attaches to both: no forward curve is used, because Deribit's dated futures on these underlyings are close to untraded — over the whole sample only three SOL and three XRP contracts produced usable daily closes. Their greeks are computed against the index instead, and the gap is documented rather than patched.

2.3 The reference asset

A fifth asset appears on the realized side only. PAXG is tokenized gold: its Deribit perpetual trades continuously like every other book here, but the underlying's price-formation market — London and COMEX — is genuinely shut from Friday 22:00 to Sunday 22:00 UTC. It therefore turns the paper's mechanism *up* rather than merely exhibiting it, and §3 uses it as a dose-response check.

It contributes no pricing estimate. Of its 5,346 listed options, 5,304 expire on a Friday, so within a trade day the weekend fraction of a contract's remaining life is a deterministic function of its maturity and cannot be separated from the maturity controls §5.1 requires. Its perpetual covers 618 days from December 2024.

3. The fact: weekends are quiet in a market that is open

Table 1 reports annualized realized volatility by day of week and the implied weekday/weekend variance ratios.

Table 1. Realized volatility by weekday, and weekend variance ratios.

	Mon	Tue	Wed	Thu	Fri	**Sat**	**Sun**	Weekend/weekday variance
BTC	59.9	58.9	61.1	60.4	59.2	**42.0**	**46.3**	**0.584**
ETH	77.0	75.9	79.2	77.1	75.3	**56.7**	**62.4**	**0.607**
SOL	89.0	85.5	83.6	82.0	87.6	**60.8**	**68.8**	**0.657**
XRP	86.5	80.1	76.5	76.0	84.2	**58.3**	**63.9**	**0.621**

Annualized percent, from five-minute returns. Weekday and weekend day counts are 2,089/834 (BTC), 1,932/772 (ETH), 655/260 (SOL) and 634/253 (XRP).

Saturday is the quietest day of the week in every asset and Sunday the second quietest, with no weekday in any row falling below any weekend day. **Weekend variance is 34% to 42% below weekday in every asset, and the exchange is open throughout.**

The agreement across assets is a sharper test of the mechanism than the level of any one ratio, and it is worth being explicit about why. If crypto weekends are quiet because the *traditional* financial system is closed, then every crypto asset should inherit the same weekly information rhythm regardless of its own microstructure, because the thing that stops on Saturday is external to all of them. If instead weekend quiet reflected asset-specific features — retail participation, venue depth, the composition of the holder base — the ratios would diverge, because those features differ sharply across these four. They agree within seven percentage points despite differences in settlement convention (inverse against linear), contract size (one against a thousand), listing history (a decade against two years) and volatility level (a factor of two).

The alternative reading, that trading itself generates variance and weekends are quiet because volume is lower, is not available here in the form it takes in equity markets: the venue is open, the order book is live, and the perpetual trades throughout. It survives only as a claim about *endogenous* participation — that traders choose not to trade at weekends — which is itself most naturally explained by there being less to trade on.

Turning the traditional calendar up. The four crypto assets all sit at the same dose of the treatment: their own venues never close, and the traditional system they take information from is shut every weekend. PAXG raises the dose. Its Deribit perpetual behaves exactly like the others, but the market that forms gold's price is genuinely shut from Friday 22:00 to Sunday 22:00 UTC, so the weekend removes not merely the flow of macroeconomic news but the price-formation process itself.

PAXG's weekend variance ratio is **0.347**, against 0.584 to 0.657 for the four crypto assets — a weekend discount of 65% where theirs is 34% to 42%, roughly double. The venue, the estimator, the sampling frequency and the day-type definition are identical; the only thing that changes is whether the underlying's own market is running. A dose-response of this shape is difficult to produce from any account that locates the effect in crypto microstructure.

This is a lower bound, and the reason matters for how it should be read. PAXG's perpetual repeats 88% of its weekend five-minute closes against 78% of its weekday ones, and stale prices depress measured variance in both regimes while biasing their *ratio* toward one. The true weekend effect in gold is therefore at least this large. Its variance ratio also falls monotonically, from 0.345 to 0.188, as the sampling interval coarsens and the staleness is removed — the signature of exactly this bias. The four traded books show no such drift, which is what licenses reading their ratios at face value.

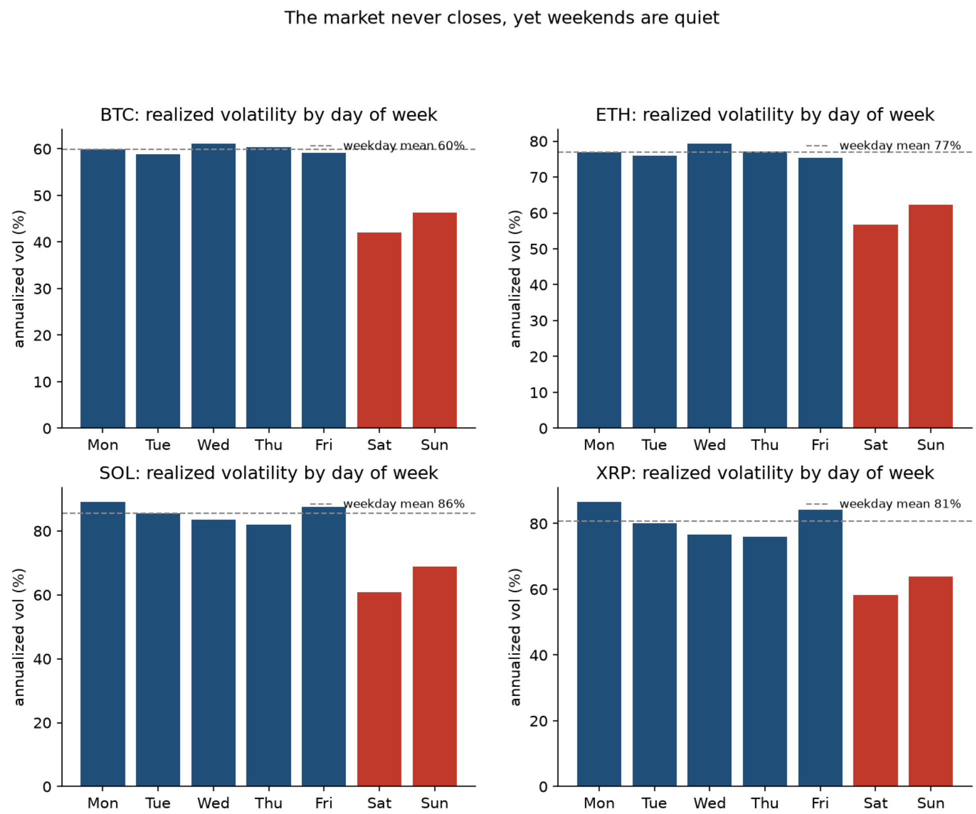

Figure 1. Realized volatility by day of week. Annualized, from five-minute returns, as a deviation from each asset's own weekday mean. Saturday and Sunday shaded.

4. Identification

Write the total variance over an option's life as a time-weighted average of a weekday and a weekend variance:

$$\sigma_{i,t}^2 T_{i,t} = v_t^{wd}(1 - w_{i,t})T_{i,t} + v_t^{we}\, w_{i,t}T_{i,t},$$

where $w_{i,t}$ is the fraction of contract i's remaining life at time t falling on a Saturday or Sunday. Dividing through,

$$\sigma_{i,t}^2 = v_t^{wd} + (v_t^{we} - v_t^{wd})\, w_{i,t}.$$

The estimating equation is

$$\sigma_{i,t}^2 = \gamma_t + \beta\, w_{i,t} + \delta' X_{i,t} + \varepsilon_{i,t},$$

with γ_t a day fixed effect and X containing log maturity, its square, and $|\Delta|$. The coefficient β estimates $v^{we} - v^{wd}$, and the implied weekend variance ratio is $(\bar{v}^{wd} + \beta)/\bar{v}^{wd}$. Standard errors are clustered by day throughout.

Two features make this credible. First, γ_t absorbs the level of volatility and everything else common to the day — the state of the market, the funding environment, any news — so β is identified only from contracts quoted **at the same instant** that differ in weekend exposure. Second, w is mechanical: it is a property of the calendar and the expiry date, not of anything the market chooses in response to volatility. A contract's weekend fraction was determined when the exchange set its listing schedule, typically years earlier.

Weekend fraction is computed exactly, in closed form, rather than by counting whole days. Deribit expiries settle at 08:00 UTC, so almost every contract covers part-days at both ends, and a whole-day count would misstate w by up to two thirds of a day on precisely the short contracts that carry most of the identifying variation. The same construction yields the fraction falling on each of the seven weekdays, which §5.2 uses.

The maturity controls deserve a word, because w and T are mechanically related: a very short contract listed on a Friday is nearly all weekend, while a long one is close to the 2/7 calendar share. If the implied volatility surface has any term structure — and it does — then omitting maturity would load part of that term structure onto β. Log maturity and its square absorb the smooth part; the identifying variation that remains is the difference between two contracts of *similar* maturity quoted at the same instant whose expiry dates fall on different weekdays. That variation exists in this venue precisely because expiries are daily.

The threat this design does not rule out by construction is a listing schedule chosen in response to expected weekend variance. We treat it as a maintained assumption. A calendar that lists an expiry on all 365 days of the year, fixed in advance and uniform across underlyings, makes it a mild one.

5. Results

5.1 The market prices most of the weekend effect

Table 2. Implied and realized weekend variance ratios.

	slope β	se	t	n	implied ratio	realized ratio	gap	(se)
BTC	−0.1612	0.0149	**−10.8**	5,339,133	0.635	0.584	+0.051	(0.066)
ETH	−0.2461	0.0277	**−8.9**	3,624,938	0.645	0.607	+0.038	(0.067)
SOL	−0.3568	0.0243	**−14.7**	189,906	0.558	0.657	−0.099	(0.102)
XRP	−0.5513	0.0691	**−8.0**	70,128	0.488	0.621	−0.133	(0.123)

Standard errors clustered by day, over 3,413, 2,703, 886 and 885 days. The gap's standard error combines the implied slope's with the realized ratio's own sampling error.

The slope is negative and overwhelmingly significant in all four: **the market does price weekend calendar time, and prices a large share of it.** Bitcoin implies a weekend variance ratio of 0.635 against a realized 0.584 — a discount of 36.5% where the true discount is 41.6% — and Ether 0.645 against 0.607. Roughly seven eighths of the effect is in the price in both established books.

The residual gaps are not individually significant, and that constraint binds everything said about them. The four t-statistics are +0.78, +0.57, −0.92 and −1.07. The implied side is estimated to within about three points of variance ratio, but the realized side is a difference of two means over at most 834 weekend days, and it dominates the uncertainty. Nothing below should be read as a measured pricing error in a particular book.

The identification is visible directly in the data, without any regression. Demeaning both squared implied volatility and weekend fraction within each trading day strips out the volatility level and every common shock, leaving only the comparison the regression uses: contracts trading at the same instant that differ in how much weekend they span. Binned that way, the relationship is close to linear across the full range of weekend fraction in all four books, which is what the constant-within-regime variance assumption behind §4 requires. (The binscatters are in the e-companion.)

The raw pattern survives an even cruder cut. Average at-the-money implied volatility for contracts with three days or less to expiry, by expiry weekday, puts Sunday expiries cheapest of the week in all four books and Monday expiries second-cheapest in three — precisely the two that carry the most weekend calendar time. The market's ordering is right; its magnitude is what §5.1 measures.

Does the market apply one discount to every underlying, or calibrate to each? Pooling the four books and testing the restrictions directly, a strictly uniform discount is rejected. Beyond that the test is close to uninformative, and saying so is more useful than reporting the non-rejection of the alternative.

The reason is visible in the dispersion. Across the four assets the standard deviation of implied weekend discounts is 0.0919 against 0.0384 for realized ones — **the market's quotes vary 2.4 times more across assets than the weekend risk they are pricing does.** That is over-differentiation rather than calibration. The proportionality hypothesis nonetheless fails to reject, and the reason is power, not support: the four realized effects are bunched inside a span of 0.092, so dividing by them is close to dividing all four slopes by the same constant, and the ratio test is then nearly the equality test. It survives only because the delta-method correction for realized-effect uncertainty inflates the standard errors on the two short samples to 0.418 and 0.502, against 0.178 and 0.187 for the mature books. A non-rejection obtained that way is a statement about the sample, not about the market.

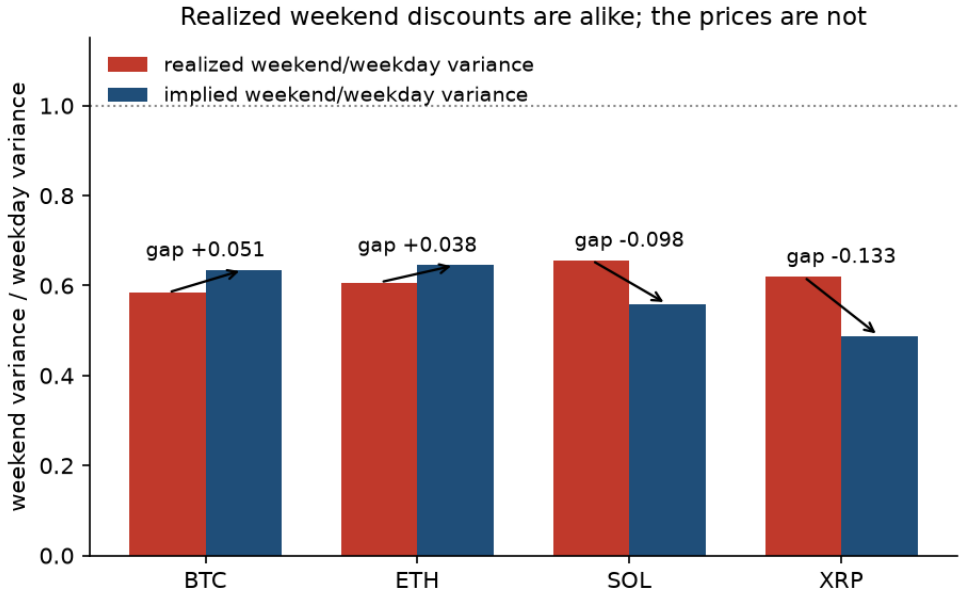

Figure 2. Implied against realized weekend variance ratios. Each asset's implied ratio with its confidence interval against its realized ratio; the 45-degree line is correct pricing.

5.2 The failure is within the weekend

Everything so far concerns the level of the weekend discount, where the market does well. Its resolution is a separate question, and the answer is sharper.

Estimating a full day-of-week profile — a separate implied effect for each of the seven days, against the realized volatility of that day — the market's ordering is otherwise good. Implied and realized profiles correlate at +0.98, +0.88, +0.89 and +0.82 across the four books. The market knows Tuesday from Thursday. One place breaks, and it breaks the same way in every book.

Table 3. Saturday against Sunday, implied and realized.

	implied Saturday effect	t	realized Saturday effect
BTC	+0.004	0.14	−0.033
ETH	+0.074	2.01	−0.094
SOL	−0.012	−0.18	−0.075
XRP	+0.158	1.21	−0.041

Effect of a Saturday relative to a Sunday, in variance-ratio units. Positive implied values mean the market prices Saturday as the busier day.

In every asset the market prices Saturday as indistinguishable from — or busier than — Sunday, when Saturday is the quieter of the two. Not one of the four implied Saturday effects is significantly negative, while all four realized effects are negative. **That is the one place in the weekly profile where the market's ranking is demonstrably wrong rather than merely compressed**, and it is the same place in all four — across two settlement conventions, a decade of listing history, and a factor of thirty in trade count.

This is a stronger form of evidence than the level results, for two reasons. Compression of a discount toward zero is what any smoothing or regularisation in a pricing model would produce, and is consistent with the market knowing the answer and shading it. Ranking two adjacent days the wrong way round is not: it requires the weekend to be represented as a single object. And because Saturday and Sunday sit inside the same weekend, an error between them cannot be attributed to anything that varies at weekly frequency — the level of volatility, the news calendar, the funding environment — all of which are common to the pair.

And it is the one error the listing schedule protects. The natural response is to spread it: sell the Saturday-heavy contract, buy the Sunday-heavy one. Both legs sit inside the weekend, so the common weekend discount differences out and what is left is exactly the failure to rank the two days. That trade cannot be put on. Expiries are daily at 08:00 UTC, so which weekday an option's remaining life falls on is fixed entirely by its entry time and expiry date; enumerating every hour of the week against every expiry in the half-day-to-eight-day window gives not a sample of the menu a desk faces but the whole of it. Monday through Friday — 120 of the week's 168 hours — **no Sunday-heavy contract is listed at all**, because any contract reaching Sunday must pass through Saturday to get there. The two legs coexist in 13 hours, on Saturday between 07:00 and 13:00 UTC and on Sunday between 15:00 and 20:00, by which time the Saturday being sold is already under way or finished.

So the market's one demonstrably wrong ranking of calendar time is also the one its own contract calendar prevents anyone from arbitraging directly. That is a more economical account of why this error survives than convention or inventory, and unlike those it needs no quote-level data — it follows from the expiry schedule alone.

5.3 A decade-long trend, and what it tracks

The implied discount is not static. **Table 4** reports it by year.

Table 4. Implied weekend variance ratio, by year.

	2020	2021	2022	2023	2024	2025	2026
BTC	0.91	0.93	0.48	0.58	0.51	0.42	**0.30**
ETH	1.00	0.95	0.55	0.58	0.45	0.49	**0.47**
SOL	—	—	—	—	0.60	0.53	0.58
XRP	—	—	—	—	0.45	0.52	0.62

In 2020 Bitcoin and Ether priced essentially no weekend discount at all, against realized ratios of 0.52 and 0.70 in the same year. By 2026 they price 0.30 and 0.47. Weighting each year by its own precision, the implied weekend effect falls by **0.144 per year in Bitcoin ($t = -12.5$) and 0.117 in Ether ($t = -10.5$)**. This also dissolves the apparent cross-sectional split in Table 2: Solana and XRP quote deeper discounts than Bitcoin and Ether largely because their samples begin in 2024, late in a market-wide trend.

Measured as this paper measures it everywhere else — mean weekend variance over mean weekday variance — the realized ratio has no significant trend over the same years ($t = -1.25$ and -1.09). That contrast looks like a market walking away from a benchmark standing still. It is not, and the reason is a measurement point worth stating in general form.

A ratio of means of a right-skewed series is a weak test, and a weak test is not a null. Daily realized variance is extreme enough that each year's mean is set by a handful of days. Three diagnostics separate an absent effect from an underpowered estimator, and all three say the same thing here.

Table 5. The weekend trend, by moment of the realized distribution.

	ratio at mid-sample	per year	t	2020 → 2026
BTC arithmetic (ratio of means)	0.494	−0.062	−1.19	0.606 → 0.402

	ratio at mid-sample	per year	t	2020 → 2026
BTC **geometric** (ratio of centres)	0.437	**−0.136**	**−7.54**	0.685 → 0.279
ETH arithmetic	0.564	−0.037	−0.97	0.637 → 0.499
ETH **geometric**	0.532	**−0.098**	**−6.25**	0.736 → 0.384

Log points per year, within-month fixed effects, month-clustered standard errors.

First, **trimming**. Cutting the top one per cent of days from *each day type within each year* — so the trim itself carries no weekend effect — takes Bitcoin's arithmetic trend from −0.062 (t = −1.2) to −0.132 (t = −4.9), and the estimate then barely moves as the trim deepens to the median. The flatness was bought entirely in the extreme right tail.

Second, **refitting at the centre**. The same contrast on log variance estimates the ratio of geometric means and carries three to four times the t-statistic on the same days. The typical Bitcoin weekend has gone from carrying 69% of a typical weekday's variance to 28%.

Third, **the sampling ladder**, which discriminates economics from measurement. Observation noise adds roughly the same amount to every day's measured variance, pushing a measured ratio toward one and attenuating any trend in it — hardest on the finest grid. A *real* trend seen through noise therefore **strengthens** as the interval coarsens; a trend manufactured by *shrinking* noise weakens. Bitcoin's geometric trend runs −0.136 at five minutes and −0.193 at two hours, and Ether's −0.098 to −0.160. It strengthens. Zero-return shares show no upward drift in either book, so growing weekend staleness is not the explanation.

Which of the two is the market pricing? Both realized measures are now falling, so the question is no longer whether the market responds to something but to which something. Fitting the implied ratio quarter by quarter gives trends of −0.186 (t = −10.43) in Bitcoin and −0.149 (t = −10.46) in Ether, against geometric realized trends of −0.136 and −0.098 and arithmetic ones of −0.062 and −0.037. Differencing implied against realized at each sampling interval, the t-statistic on the difference against the geometric benchmark falls monotonically to +0.2 and +0.4 at the coarsest sampling, where microstructure noise cannot reach; against the arithmetic benchmark it sits near two standard errors throughout and closes only where power runs out. **The implied trend converges on the geometric benchmark and never on the arithmetic one.**

A backward-looking level test agrees. Regressing each quarter's log implied ratio on the *previous* quarter's realized ratio with asset fixed effects — which is what calibrating to recent history means — gives slopes of 0.71 on the geometric ratio and 0.27 on the arithmetic one. Both regressors are estimates, so both coefficients are attenuated; correcting each by its own sampling variance lifts them to 1.15 and 0.48. The correction is large enough that only the ranking should be read off it, and the ranking is the one the trends give: the market's quotes move roughly one-for-one with where the weekend's variance usually sits, and about half as much with where its mean sits.

An option, however, pays off on expected total variance — an arithmetic mean. **The market is tracking a real decline, at close to the right speed, in the wrong moment of the distribution.** §6 shows that no proportional risk premium can move a variance ratio, so the mismatch is not repaired under the pricing measure. Its size is the gap between a weekend's typical variance and its average one, and that gap has been widening: measured as log mean minus mean log, a scale-free index of right-tail weight, the weekend's excess over the weekday's runs from −0.23 in 2020 to +0.31 in 2026 for Bitcoin and −0.07 to +0.27 for Ether. The weekend has not simply gone quiet; it has become *lumpier* — ordinary weekends much quieter, rare weekends no less violent. A market that watches the middle of the distribution will mark the weekend down faster than its mean falls, by precisely the amount the tail is growing.

Two limits on this, stated rather than buried. Solana and XRP show no trend on either side, which is consistent with a market-wide arc that began before they listed, but their standard errors are three to thirteen times the mature books' and their implied trends rest on nine or ten quarters; no difference test rejects anything for either, so they neither confirm the mechanism nor contradict it. And the comparison treats the option side and the bar side as independent when a volatile week moves both; the dependence is positive, which makes the reported standard errors conservative for the null that the two trends are equal and anti-conservative for the null that they differ.

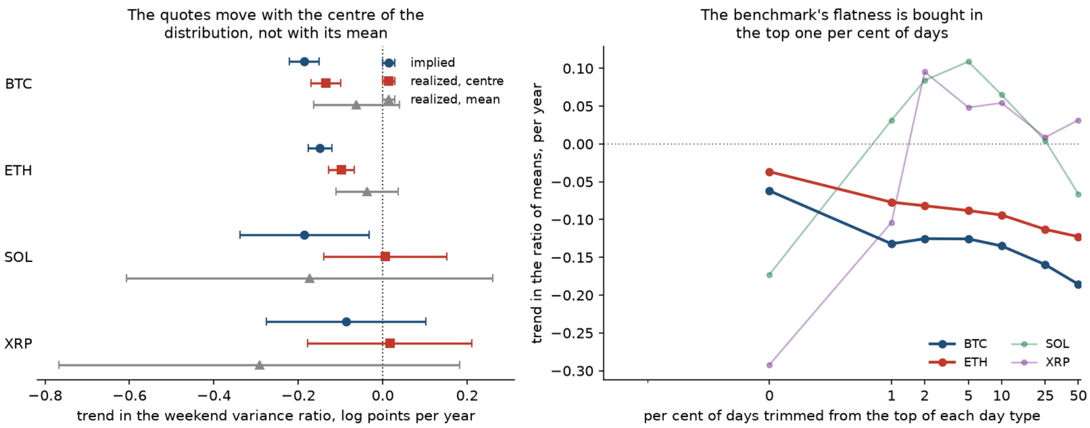

Figure 3. What the market tracks. Left: implied, realized-centre and realized-mean trends with 95% intervals, per asset. Right: the realized trend as the trimming fraction deepens.

6. Racing the risk-based explanation

Implied and realized quantities are measured under different probability measures, so a market pricing the weekend above its physical variance could be charging a risk premium rather than making an error. We take that seriously enough to race it rather than dismiss it, and two features of the setting constrain what a premium can do.

A proportional premium cancels. Suppose the risk-neutral measure inflates variance by a factor λ applied uniformly to calendar time. Then both the weekend and the weekday numerator and denominator scale by λ, and the *ratio* we measure is unchanged. Only weekend-*specific* risk — something about Saturday and Sunday that differs in kind, not merely in level — can move the quantity in Table 2. This is the constraint that makes the ratio formulation worth the loss of information relative to levels.

Weekend tails are genuinely fatter. Standardized for scale, weekend returns carry 15% to 28% more mass beyond five standard deviations than weekday returns in all four books. So weekend-specific risk exists and has to be priced explicitly rather than argued away. (A companion tail-asymmetry result on skew does *not* replicate — two of the four books move the wrong way — so the race below runs on tail mass only.)

Bounding what a jump premium can buy. Split realized variance in each regime into a continuous part and a jump part using truncated realized variance, and let the risk-neutral measure price jump variance at a multiple $\kappa \geq 1$ of its physical value. The variance a dealer would charge is then the continuous part plus κ times the jump part, and the priced weekend ratio $R^*(\kappa)$ is monotone in κ. Its limit as $\kappa o\infty$ is therefore a **bound**: no jump-risk premium of any size can push the priced ratio past it.

The split uses a three-standard-deviation truncation with the local scale taken from the same day's bipower variation, so a quieter weekend is not mechanically classified as jump-free, and with an intraday seasonality factor estimated separately for each regime. Truncation misclassifies roughly 0.3% of ordinary returns by construction; the bias applies to both regimes and largely cancels in the ratio. That the measured jump shares are not an artefact of the estimator's floor is checked directly: run on a simulated pure diffusion of the same length and frequency, the same estimator returns a jump share under 5%, against the 24% to 36% these series produce.

Table 6. Residual gap under a jump-risk premium of size κ.

	κ = 1	κ = 3	κ = 10	bound ($\kappa o \infty$)	reachable?
BTC	+0.051	+0.034	+0.019	**+0.009**	no
ETH	+0.039	+0.048	+0.056	**+0.039**	no
SOL	−0.098	−0.139	−0.184	**−0.098**	no
XRP	−0.136	−0.144	−0.150	**−0.136**	no

Implied ratio minus the ratio a market pricing weekend jump variance at κ times physical would quote. "Reachable" asks whether any κ closes the gap.

Jump compensation cannot account for the gap in any of the four books, and it fails in three distinct ways. Bitcoin is the near miss: a premium does push its priced ratio toward the quote, but the whole path from $\kappa = 1$ to the bound closes only 82% of a gap of 0.051, and it takes an unbounded premium to get that far. Ether fails on *direction* — its weekend is less jump-intensive than its weekdays, so every increase in κ widens the gap, from +0.039 at the physical measure to +0.056 at ten times it. Solana and XRP fail on *sign*: their quotes sit below their realized ratios, while a jump premium can only push a priced ratio up. For those two not even $\kappa = 0$ works — pricing weekend jump variance at zero still leaves continuous ratios of 0.616 and 0.613 against quotes of 0.558 and 0.488. Closing their gaps needs a negative price of jump risk, which is to say a market that pays to hold it.

Sampling uncertainty softens the verdict for any single asset without rescuing the story. Block-bootstrapping whole weeks and redrawing the implied ratio from its own sampling distribution, the probability that a jump premium of some size could reach the market's quote is 0.27 for Bitcoin, 0.13 for Ether, 0.10 for Solana and 0.11 for XRP. No single one of those rejects at conventional levels. **What is decisive is that the failure repeats in four books and requires opposite-signed premia in the two pairs** — a large positive price of weekend jump risk in the mature books and a negative one in the two listed in 2024. No single risk premium does both. The result survives 44 estimator settings varying the truncation threshold, the seasonality treatment and the sampling frequency.

The option surface agrees, and disagrees with the premium story in the same direction. If weekend jump risk were being compensated, the compensation should be concentrated where jump risk is priced — in the far wings. Estimating the weekend discount separately across the moneyness ladder, the far wing discounts the weekend *harder* than the money in all four books, by 0.10 to 0.13. Fitted jointly so the contrast carries a covariance, the far-wing-minus-at-the-money difference is −0.121 (t = −4.15) for Bitcoin and −0.118 (t = −4.95) for Ether, with equality across the smile rejected outright in both ($\chi^2(3)$ = 18.4 and 28.4, $p < 0.001$). The two small books have the same sign and no precision. The at-the-money estimates also reproduce Table 2 almost exactly — 0.630 against 0.635 for Bitcoin, 0.639 against 0.645 for Ether — which confirms the widened sample has not changed the object being measured.

That the wings discount the weekend harder rather than more softly is the opposite of what weekend jump compensation implies, and it is better read as the smile's moneyness metric following the weekend clock only partway: a level adjustment made, and a structural one not.

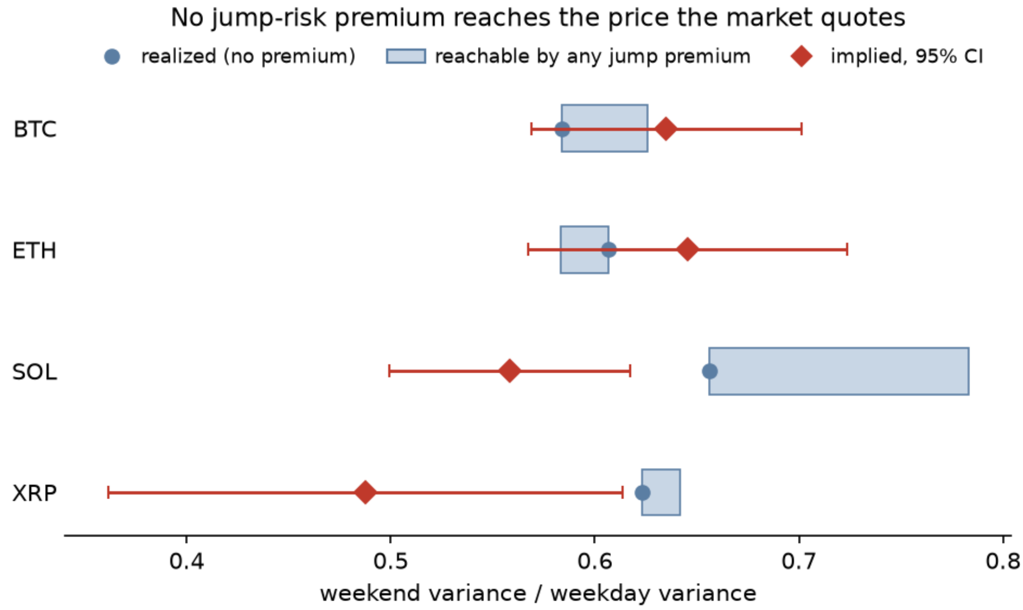

Figure 4. The horse race. Priced weekend variance ratio as the jump-risk premium κ grows, against each book's quote (diamond). The dashed line is the bound as $\kappa o \infty$. Bitcoin approaches its quote and stops short; Ether moves away from it.

7. The error is measurable and has never been harvestable

A pricing error that no one can trade against invites the objection that it is an artefact of the estimator. This section answers that objection in two steps: the error does show up in a hedged position, with the right sign and the right cross-sectional order, and it has nonetheless never cleared the venue's costs.

The trade that isolates the effect is a vega-matched calendar spread: sell the weekend-heavy contract, buy the weekday-heavy one, delta-hedge both in the perpetual, hold to settlement. Buckets are assigned within each day's own cross-section, because an absolute weekend-coverage threshold describes a spread whose two legs rarely exist at the same time. Costs are measured rather than assumed — Deribit's 0.03% option fee capped at 12.5% of premium, 0.05% taker on every perpetual rebalance, and an effective half-spread recovered from the tape by differencing buyer-paid against seller-received implied volatility on the same instrument-day.

Table 7. The spread, gross and net of measured costs (daily rehedging).

per unit vega	BTC	ETH
gross spread	+0.0418	+0.0418
exchange and hedging fees, both legs	−0.0579	−0.0581
spread crossing, both legs	−0.0083	−0.0127
taker net	**−0.0245** (t = −2.09)	**−0.0289** (t = −1.82)
maker net (earns the spread rather than paying it)	−0.0079	−0.0036

The gross P&L validates the measurement chain. Ordering the four assets by the size of their measured pricing gap orders them by gross trading P&L as well, without exception: Bitcoin's +0.051 gap and +0.023 gross, down to XRP's −0.133 and −0.025. This is a stronger check than a single profitable backtest, because the two legs differ in maturity as well as weekend coverage — a mechanical bias in which short-dated contracts out-earn longer-dated ones would generate same-signed P&L in every asset regardless of its pricing gap. Instead the sign and the rank track the gap across four books with two settlement conventions.

The signal is nonetheless smaller than the toll. Of the 0.066 per unit vega a Bitcoin taker pays, the bid-ask accounts for 0.008 and the remaining 0.058 is exchange and hedging fees, which fall on a maker exactly as they fall on a taker. Quoting rather than crossing is worth four half-spreads and closes roughly two thirds of the gap without reaching zero. Only a maker on a fee tier discounting option fees by 27% clears zero, at roughly +0.022 per unit vega — and not over the last twelve months, in which every construction we test loses money.

The pricing error cannot be harvested by the trade that documents it, and the obstacle is the venue's fee schedule rather than the market. That is consistent with the reading suggested by §5.3: a desk that prices the weekend clock correctly improves its marks on inventory it holds anyway, at no crossing cost and no capacity limit, and would leave the quoted gap largely intact while making it unavailable to anyone who has to cross to reach it. We state that as an inference, not a measurement — separating it from the alternatives requires quote-level data this tape does not carry.

8. Conclusion

Weekend variance in continuously traded crypto is 34% to 42% below weekday variance, on a venue that never closes, and roughly 65% below for a tokenized asset whose own underlying market is shut. Trading is possible throughout, so this isolates the information-arrival channel from the mechanical effect of closure — the separation French and Roll (1986) could not make, and which Barclay, Litzenberger and Warner (1990) approached from the opposite direction by opening an exchange on Saturdays. Both designs point to trading time rather than calendar time as the relevant clock, and this one adds a dose-response: raise the amount of the traditional calendar that is switched off, and the weekend deepens on the same exchange with the same estimator.

The options market prices most of it. Roughly seven eighths of the realized weekend effect is in the price in both mature books, identified from contracts quoted at the same instant that differ only in how much weekend they span. That is a substantially better performance than the practitioner literature on calendar-time conventions would suggest, and it is worth stating plainly before the failures: this market reads the calendar well.

What it gets wrong is not the level but the resolution. All four books treat the weekend as one undifferentiated block, pricing Saturday as indistinguishable from Sunday when Saturday is reliably the quieter of the two — and because the two days sit inside the same weekend, that error cannot be attributed to anything varying at weekly frequency. The venue's own expiry schedule then makes this specific error unarbitrageable, offering both legs of the correcting spread in 13 hours of the week and none of them on a weekday. **The market's one demonstrably wrong ranking of calendar time is the one its own contract calendar protects.** That is a more economical account of why an error persists than convention or inventory, and unlike those it follows from the listing schedule alone.

The implied discount has meanwhile deepened for a decade, at 0.144 and 0.117 per year. That trend turns out to be a response to a real change in the data measured against the wrong moment of it. The typical weekend really has been getting quieter — 0.136 log points a year at the centre of Bitcoin's distribution, against an arithmetic benchmark that appears flat only because each year's mean is set by a handful of days. The market tracks the centre; an option pays off on the mean. Since §6 rules out any proportional risk premium moving a variance ratio, the mismatch is not repaired under the pricing measure, and its size is the widening gap between a weekend's typical variance and its average one.

That last result carries a methodological point beyond this application. A ratio of means of a right-skewed series is a weak estimator, and a weak estimator returning nothing is not evidence that there is nothing there. Three cheap diagnostics separate the two — trim the tail and refit, refit at the centre of the distribution, and walk the sampling interval — and here all three agree that a non-rejection was a power failure. Where a non-rejection is load-bearing, it warrants the same scrutiny a rejection would receive.

Three limitations bound these claims. Solana and XRP have standard errors three to thirteen times the mature books', so they replicate the central realized fact without sharpening any of the pricing tests, and we have been careful not to lean on them. The gold result is a lower bound, because PAXG's weekend staleness biases its measured ratio toward one. And the identification assumes the expiry schedule is exogenous to weekend variance — a maintained assumption, though a listing calendar fixed years in advance and uniform across underlyings makes it a mild one.

What we cannot say from a trade tape is *why* a desk would calibrate to the centre of a right-skewed distribution when the payoff is on its mean. Robust estimation of a skewed quantity is ordinary practice and defensible on its own terms; it is simply the wrong practice for this quantity. Separating deliberate robustness from a fitted model with a thin-tailed error, or from nobody looking, requires quote-level data and a view of who is on the other side of the trade.

Electronic companion

The e-companion contains: the full identification appendix and within-day binscatters for all four assets; the pooled convention test and its power analysis; robustness of the realized effect across sampling frequencies, jump truncation thresholds and seasonality treatments; the complete day-of-week profile tables; the full trimming and sampling ladders behind Table 5; the 44-setting robustness grid behind Table 6; the smile and wing analysis; and the complete trading results, including the rehedge ladder, the outright-short variant, the contract-selection sweep and the maker fee decomposition. All code and data-construction scripts are in the project repository, together with the test suite that pins every estimator reported here.